

Machine Overview, Technical Specifications & System Architecture

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2. Machine Overview

The EquipmentIQ VMC-3000 is a three-axis vertical machining centre designed for high-precision milling, drilling, boring, and tapping operations on aluminium, steel, and titanium alloys. The machine incorporates an integrated condition monitoring system comprising tri-axial accelerometers (Bosch CISS sensor, 2 kHz sampling), thermal sensors, and drive-level current monitoring, providing real-time health data to the EquipmentIQ monitoring platform.

The VMC-3000 is configured for fleet deployment across three machine instances (M01, M02, M03) in a brownfield production environment. Sensor data is timestamped and transmitted via OPC-UA to the EquipmentIQ edge gateway at 2,000 samples per second per axis.

NOTE: This document is the primary reference for service engineers commissioning, troubleshooting, or performing preventive maintenance on VMC-3000 series machines.

3. Technical Specifications

Parameter	Specification
Machine Type	Vertical Machining Centre, 3-axis (X/Y/Z)
X-Axis Travel	800 mm
Y-Axis Travel	500 mm
Z-Axis Travel	600 mm
Table Size	900 mm × 500 mm
Table Load Capacity	600 kg
Spindle Speed Range	100 – 8,000 RPM (standard); 12,000 RPM optional
Spindle Taper	ISO 40 / BT40
Spindle Power (rated)	18.5 kW (S1) / 22 kW (S6-25%)
Spindle Torque (max)	145 Nm
Rapid Traverse (X/Y/Z)	36 / 36 / 30 m/min
Feed Rate Range	1 – 10,000 mm/min
Positioning Accuracy	±0.005 mm (per ISO 230-2)
Repeatability	±0.003 mm
Tool Magazine Capacity	30 tools (arm-type ATC)
Max Tool Diameter	80 mm (adjacent pockets empty: 150 mm)
Max Tool Length	300 mm
Max Tool Weight	8 kg
ATC Change Time	3.5 s (chip-to-chip)

Table 3-1: Mechanical Specifications — VMC-3000

Parameter	Specification
Supply Voltage	3-phase, 380–420 V AC, 50/60 Hz
Power Consumption (rated)	35 kVA
DC Bus Voltage	560–620 V DC (nominal 590 V)
Main Fuse Rating	80 A (gl/gG type)
Ground Leakage (max)	30 mA (RCD trip threshold)
UPS Battery Backup	20 min (CNC memory protection)
Compressed Air Supply	6 bar, 200 L/min, ISO 8573-1 Class 3
Coolant Pump Power	2.2 kW
Lubrication Pump Power	0.18 kW
Control Voltage	24 V DC (internal)
CNC Controller	EquipmentIQ CNC-500 (Fanuc 31i-B compatible)

Table 3-2: Electrical & Utility Specifications — VMC-3000

4. System Architecture

The VMC-3000 system architecture comprises ten integrated subsystems, each monitored by dedicated sensors and reporting to the CNC controller via CANopen fieldbus (ISO 11898). The Module Identification (MID) numbering follows SAE J2012 convention.

MID	Subsystem Code	Subsystem Name	Primary Sensors	Bus
128	SPN	Spindle Drive System	Accelerometer, Temp, Current	CANopen
130	AXS	Axis Servo System	Accelerometer, Temp, Current	CANopen
132	TCS	Tool Change System	All sensors	CANopen
134	CLS	Coolant System	Pressure, Flow, Level	CANopen
136	LUB	Lubrication System	Pressure, Flow, Level	CANopen
138	HYD	Hydraulic System	Pressure, Flow, Level	CANopen
140	CNC	CNC Controller	CPU, Memory, Bus	CANopen
142	ELC	Electrical Cabinet	Voltage, Current, Temp	CANopen
144	VIB	Vibration Monitoring	Accelerometer, Temp, Current	CANopen
146	THM	Thermal Management	Voltage, Current, Temp	CANopen

Table 4-1: Subsystem Architecture — MID Mapping

5. Subsystem Descriptions

Spindle Drive System (SPN, MID 128)

The spindle system delivers 18.5 kW continuous power via a 3-phase induction motor with integral encoder. Bearing health is monitored by the Bosch CISS tri-axial accelerometer mounted at the spindle nose, sampling at 2,000 Hz. Vibration data is processed on the EquipmentIQ edge gateway to compute RMS, crest factor, and kurtosis index in real time. Refer to DOC-EIQ-002 for full spindle service documentation.

Axis Servo System (AXS, MID 130)

Three linear axes (X, Y, Z) driven by AC servo motors with absolute encoders (17-bit resolution). Ball screws (grade C3) and linear guides (grade H) provide positioning accuracy of ± 0.005 mm. Servo drives communicate with the CNC controller via FSSB (Fibre-optic Servo Serial Bus). Refer to DOC-EIQ-003.

Tool Change System (TCS, MID 132)

A 30-position arm-type automatic tool changer (ATC) provides chip-to-chip change time of 3.5 s. Tool presence and identity are monitored by inductive sensors in each pocket. Tool length is verified by in-process touch probe after each change. Clamp force is monitored by pressure transducer (P021, normal 55–75 bar).

Coolant System (CLS, MID 134)

The coolant system delivers up to 40 L/min through-spindle and flood coolant. A 2.2 kW centrifugal pump draws from a 300 L stainless tank. Coolant temperature is controlled by an air-cooled chiller (setpoint $20\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$). Concentration and flow are monitored continuously. Refer to DOC-EIQ-004.

Lubrication System (LUB, MID 136)

A centralised oil-pulse lubrication system (Bijur Delimon type) delivers measured oil volumes to all linear guide rails and ball screw nuts on a timed cycle (default: 15 min interval, 2 s pulse). Oil grade: ISO VG 32 way lube oil. Refer to DOC-EIQ-004.

Hydraulic System (HYD, MID 138)

A dedicated hydraulic unit (8 L/min, 70 bar nominal) supplies the workpiece clamping fixtures and ATC arm swing mechanism. System pressure is monitored by P050 (normal 60–80 bar). Oil temperature is controlled below 55 °C.

CNC Controller (CNC, MID 140)

The EquipmentIQ CNC-500 controller runs a real-time Linux kernel with a Fanuc 31i-B compatible NC interpreter. 10 MB of NC program memory, 64-axis interpolation capability, and OPC-UA server for data export. Refer to DOC-EIQ-006.

Electrical Cabinet (ELC, MID 142)

The IP54-rated electrical cabinet houses the main contactor, servo drives, DC bus, UPS module, and PLC I/O racks. Cabinet temperature is maintained below 45 °C by forced-air cooling with filtered intake. Refer to DOC-EIQ-006.

Vibration Monitoring (VIB, MID 144)

The Bosch CISS tri-axial MEMS accelerometer ($\pm 8g$ range, 2 kHz) is permanently mounted at the spindle nose. Vibration severity is classified per ISO 10816-3 Zones A–D. Refer to DOC-EIQ-005.

Thermal Management (THM, MID 146)

Eight NTC thermistors monitor spindle motor, servo drives, electrical cabinet, ambient, coolant, hydraulic oil, and feed motor temperatures. Thermal compensation for axis positioning error is active above 25 °C ambient.

6. Nameplate & Identification

The machine nameplate is located on the rear of the machine column, above the electrical cabinet. The following information is stamped on the nameplate:

Field	Example Value	Description
Model	VMC-3000-B	Machine model and variant
Serial Number	EIQ-M01-2019-0042	Unique machine identifier (Machine ID for monitoring)
Manufacturing Date	2019-02	Year-Month of manufacture
Spindle Power	18.5/22 kW	Rated / peak power
Supply Voltage	3~ 400V 50Hz	Electrical supply specification
IP Rating	IP54	Enclosure protection class
CE Mark	CE 2018/037	Conformity declaration reference

Table 6-1: Nameplate Fields

NOTE: The Serial Number maps directly to the Machine ID used in the EquipmentIQ monitoring platform (M01, M02, M03 in the Bosch CNC dataset). Always quote the full serial number when logging a service request or complaint.

7. Installation Requirements

- Floor loading: minimum 6,000 kg/m2 reinforced concrete slab, 300 mm minimum thickness.
- Vibration isolation: anti-vibration levelling pads (supplied), torque to 80 Nm after 48-hour settling.
- Supply cable: 4-core 16 mm2 Cu, max 20 m from distribution panel to machine isolator.
- Earth bonding: dedicated PE conductor, max 1 ohm machine-to-earth resistance.
- Compressed air: 8 mm push-fit connector, 6.0–7.0 bar, dry and filtered (ISO 8573-1 Class 3).
- Coolant drain: DN50 gravity drain within 1 m of machine base.
- Ambient temperature: 5–40 °C operating, 40–75 % RH non-condensing.
- Lighting: minimum 500 lux at machine working area (EN 12464-1).

8. Safety Summary

DANGER: Before performing any maintenance on the VMC-3000, complete a full Lockout/Tagout (LOTO) procedure on the main isolator. Verify absence of voltage with a calibrated tester. The machine stores hazardous energy in the DC bus capacitors for up to 5 minutes after power-off.

WARNING: The spindle generates significant force during tool change. Never position hands near the ATC arm during an active or partially-completed tool change cycle.

- Emergency stop buttons are located on the operator panel, rear of machine, and electrical cabinet door.
- The machine is guarded to EN ISO 23125 (machine tools – safety – turning machines). Never bypass interlocks.
- Coolant and lubricant disposal must comply with local environmental regulations.

- Use correct PPE: safety glasses, steel-toed boots, cut-resistant gloves when handling tools.
- Refer to SDS sheets for coolant concentrate (P031) and way oil (P040) before handling.